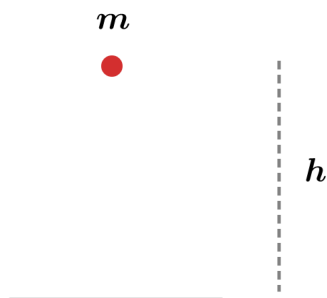


2019B F=ma Exam: Problem 22

Kevin S. Huang



By conservation of energy,

$$\frac{1}{2}mv^2 = mgh$$

so the velocity of the apple right before hitting the ground is

$$v = \sqrt{2gh}$$

By the impulse-momentum theorem,

$$Ft = \Delta p = mv = m\sqrt{2gh}$$

The average pressure exerted on the apple is

$$P = \frac{F}{A} = \frac{m\sqrt{2gh}}{At}$$

Thus,

$$P_{min} = \frac{(0.2 \text{ kg})\sqrt{2(10 \text{ m/s}^2)(0.1 \text{ m})}}{(0.0004 \text{ m}^2)(0.1 \text{ s})} = 7100 \text{ Pa}$$

so the answer is $\boxed{\text{C}}$.